

Questionnaire Rule Engine Guide

Architecture

Rules live in `config/questionnaire_rules.php`. Laravel sends them to the frontend `QuestionnaireEngine`. The engine processes questions using `question_key`, never question text.

Current Actions

`filter`: keep only mapped options based on a previous answer.
`hide_option`: remove specific options when a condition matches.
`skip_question`: completely skip a question when a condition matches.

Filter Example

`source=country, target=sub_region, action=filter, mapping: France -> [Loire Valley, Bordeaux]`. JS: `filterOptions()` keeps only mapped options.

Hide Option Example

`source=wine_type, target=sub_region, action=hide_option, when=Red, options=[Champagne]`. JS: `applyHideOptionRule()` removes Champagne.

Skip Question Example

`source=wine_type, target=tannin, action=skip_question, when=White`. JS: `shouldSkipQuestion()` returns true.

Future Actions

`show_question, set_default, clear_answer, require, disable_option, sort_options, message`.

Adding New Rules

If using an existing action, only edit `questionnaire_rules.php`. For a brand-new action, add a new case inside `filterOptions()` (or another dedicated engine method) and implement the behavior.

Best Practices

Use `question_key`. Keep rules in config, not JS. Match DB values exactly. Keep JS generic.